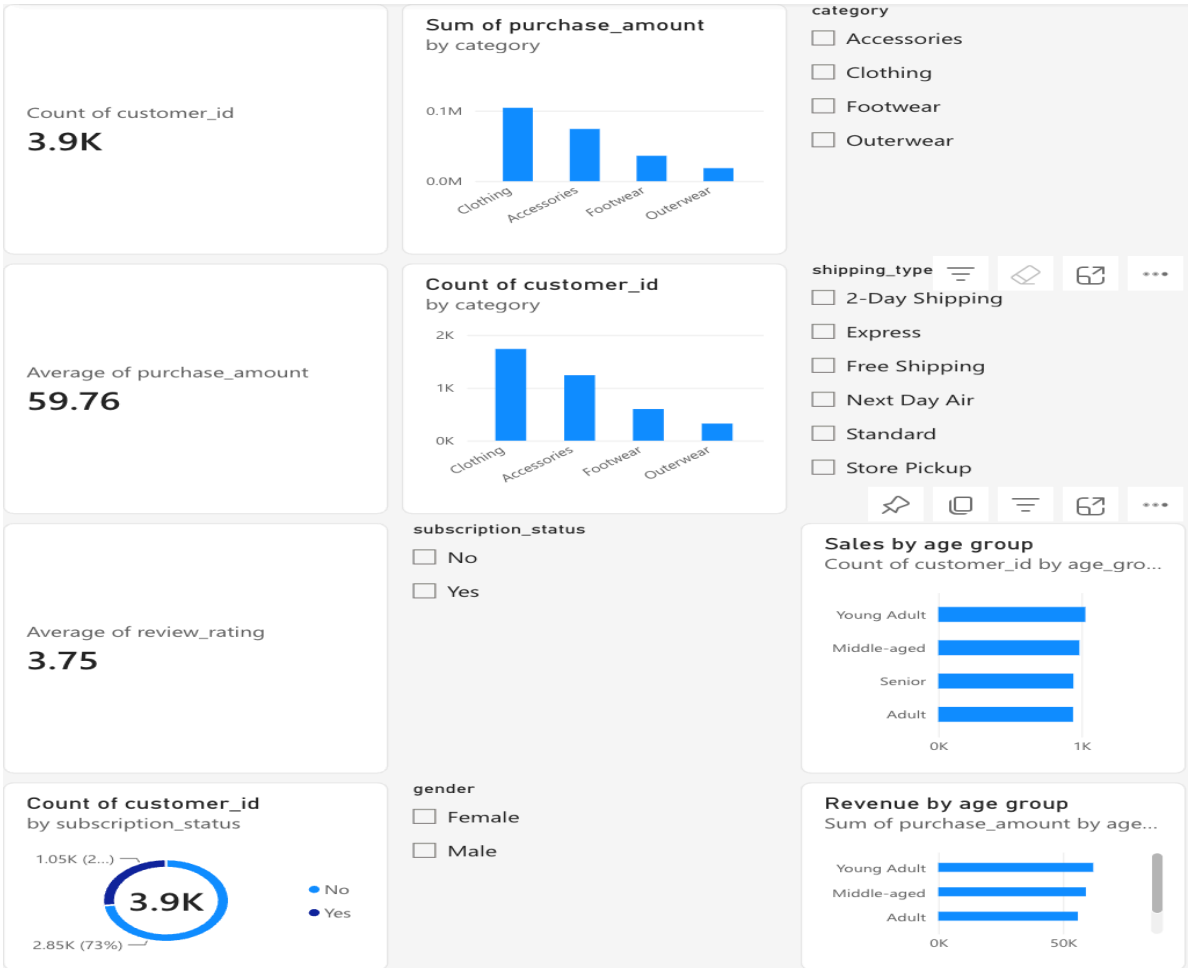
Customer Shopping Behaviour Analysis

Data: 3,900 customer purchase records
Tools: Python, PostgreSQL, Power BI

Summary

Customers spent $233,081 in total. The average purchase is $59.76, and the average review rating is 3.75 out of 5. About 27% of customers are subscribed and 73% are not.

Spending per purchase is similar for men and women, for subscribers and non-subscribers, and across age groups. The useful differences are who buys most often and which product categories bring



in the revenue.

Findings

Figure 1. Revenue by gender :Male customers generated $157,890 in revenue, compared with $75,191 from female customers. Men account for most of the revenue because

```
customer_behavior=# SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
 gender | revenue
--------+---------
 Female |   75191
 Male   |  157890
(2 rows)
```

Figure 2. Discounted purchases above the average spend:These customers used a discount and still spent at least the average purchase amount of $59.76. The amounts shown, from $64 to $97, show that a discount does not always mean a small purchase. The full result continues beyond the rows in this screenshot.

```
customer_behavior=# SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
 customer_id | purchase_amount
-------------+-----------------
           2 |              64
           3 |              73
           4 |              90
           7 |              85
           9 |              97
          12 |              68
          13 |              72
          16 |              81
```

Figure 3. Five highest-rated products

Gloves have the highest average rating at 3.86, followed by sandals (3.84), boots and hats (3.81), and T-shirts (3.78). These ratings are close to the overall average of 3.75, so no product stands far above the rest.

```
customer_behavior=# SELECT item_purchased,
       ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
 item_purchased | Average Product Rating
----------------+------------------------
 Gloves         |                   3.86
 Sandals        |                   3.84
 Boots          |                   3.81
 Hat            |                   3.81
 T-shirt        |                   3.78
(5 rows)
```

Figure 4. Average purchase amount by shipping type

Express shipping has a higher average purchase amount, $60.48, than Standard shipping, $58.46. The difference is only $2.02, so shipping type is not a strong driver of spending.

```
customer_behavior=# SELECT shipping_type,
       ROUND(AVG(purchase_amount), 2)
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
 shipping_type | round
---------------+-------
 Standard      | 58.46
 Express       | 60.48
(2 rows)
```

Figure 5. Customers, average spend, and revenue by subscription status

Non-subscribers account for most of the revenue: 2,847 customers and $170,436, compared with 1,053 subscribers and $62,645. Average spend is almost the same ($59.87 without a subscription, $59.49 with one), so a subscription is not linked to higher spending.

```
customer_behavior=# SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount), 2) AS avg_spend,
       ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue DESC;
 subscription_status | total_customers | avg_spend | total_revenue
---------------------+-----------------+-----------+---------------
 No                  |            2847 |     59.87 |     170436.00
 Yes                 |            1053 |     59.49 |      62645.00
(2 rows)
```

Figure 6. Top five products by discount rate

Hats are discounted on 50.00% of purchases, followed by sneakers (49.66%), coats (49.07%), sweaters (48.17%), and pants (47.37%). The rates sit close together, so no single product is discounted far more often than the others.

```
customer_behavior=# SELECT item_purchased,
       ROUND(
           100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)
           / COUNT(*),
           2
       ) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
 item_purchased | discount_rate
----------------+---------------
 Hat            |         50.00
 Sneakers       |         49.66
 Coat           |         49.07
 Sweater        |         48.17
 Pants          |         47.37
(5 rows)
```

Figure 7. Customers by purchase segment

Most customers are Loyal: 3,116 have more than 10 previous purchases. Returning customers (2–10 previous purchases) number 701, and only 83 are New (exactly 1 previous purchase). The base is mostly repeat buyers, not first-time customers.

```
customer_behavior=#
customer_behavior=# WITH customer_type AS (
    SELECT customer_id,
           CASE
               WHEN previous_purchases = 1 THEN 'New'
               WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
               ELSE 'Loyal'
           END AS customer_segment
    FROM customer
)
SELECT customer_segment,
       COUNT(*) AS "Number of Customers"
FROM customer_type
GROUP BY customer_segment;
 customer_segment | Number of Customers
------------------+---------------------
 Loyal            |                3116
 New              |                  83
 Returning        |                 701
(3 rows)
```

Figure 8. Top products in each category

Jewelry leads Accessories with 171 orders, ahead of sunglasses and belts (161 each). In Clothing, blouses and pants are tied at 171, with shirts at 169. In Footwear, sandals lead with 160 orders and shoes follow with 150.

```
customer_behavior=#
customer_behavior=# WITH item_counts AS (
    SELECT category,
           item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (
               PARTITION BY category
               ORDER BY COUNT(customer_id) DESC
           ) AS item_rank
    FROM customer
    GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
 item_rank |  category   | item_purchased | total_orders
-----------+-------------+----------------+--------------
         1 | Accessories | Jewelry        |          171
         2 | Accessories | Sunglasses     |          161
         3 | Accessories | Belt           |          161
         1 | Clothing    | Blouse         |          171
         2 | Clothing    | Pants          |          171
         3 | Clothing    | Shirt          |          169
         1 | Footwear    | Sandals        |          160
         2 | Footwear    | Shoes          |          150
```

Figure 9. Repeat buyers by subscription status

Among customers with more than 5 previous purchases, 2,518 are not subscribed and 958 are. Most repeat buyers are outside the subscription, so a subscription is not what keeps them buying.

Figure 10. Revenue by age group

Young Adults bring in the most revenue at $62,143, followed by Middle-aged ($59,197), Adult ($55,978), and Senior ($55,763). The four groups are close, so no single age group dominates sales.

```
         2 | Footwear    | Shoes           |          150
customer_behavior=# SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;
 subscription_status | repeat_buyers
---------------------+---------------
 No                  |          2518
 Yes                 |           958
(2 rows)

customer_behavior=# SELECT age_group,
       SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
  age_group  | total_revenue
-------------+---------------
 Young Adult |         62143
 Middle-aged |         59197
 Adult       |         55978
 Senior      |         55763
(4 rows)
```

Recommendations

1. Treat clothing as the core range. It produces nearly half of revenue. Jewelry, blouses, pants, shirts, and sandals are the products to keep in stock.
2. Do not expect the current subscription to raise spend. Subscribers and non-subscribers buy at almost the same amount. Most repeat buyers are not subscribed, so the offer needs a clearer benefit.
3. Review the discount field before acting on it. Discounts appear only on male purchases. Until that is confirmed as a real business rule, do not use it to compare men and women.
4. Keep age and shipping as secondary views. Revenue is spread across age groups, and Express shipping is only slightly above Standard.